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How can EDA help solve challenges in automotive connectivity?

Connected vehicles are one of the most challenging parts in automotive, as they require data processing, Ethernet connectivity, antennae and more connections. The only way engineers can do modern designs in wiring these together is through integration between mechanical and electrical software, which may involve electrical analysis and connectivity issues. Can modern EDA help you here?

Walden C. Rhines, chairman and CEO of Mentor Graphics, speaks to Dilin Anand and Jitin David of EFY

What is the biggest new challenge solved by EDA tools for the automotive industry?

Complexity has grown so fast that they cannot manage it in any other way. Especially, now there are safety, regulatory and performance requirements that just make it absolutely imperative that they verify their designs for all sorts of conditions. You need to show a paper trail that you have verified the operations over all extremes of temperature, analysed the electrical characteristics thoroughly, anticipated the impact of various failures that could occur and ensured that they were not catastrophic. The only way you can verify these is virtually.

Comparing with challenges from a decade ago, how have automotive electronics engineers' need for EDA tools evolved?

Since introducing the first wire harness product in 1992, it has been a fairly slow evolution. However, the good news for us is that automotive electronics has increased to a level where it is no longer possible to design wiring by hand. You just have to use automated/semi-automated wiring software—it is very much like the placing problem in printed circuit board (PCB) or integrated circuit. The engineer has constraints due to all those rules that automotive companies accumulated, and then there are mechanical constraints too.

Could you share an instance of the new kind of constraints that these software help tackle?

The software may have constraints such as 'no more than six wires in a bundle in a wet zone of a car,' and 'no more than ten wires in the dry zone,' or it may have a con-

straint that says, 'be sure that all the wire bundles can fit through the hole in the door frame,' all of which you have to verify without physically trying to force a bundle of wires through the hole each time.

How does software help solve this problem?

The only way you can do this is through integration between mechanical and electrical software, which may involve problems such as electrical analysis and connectivity issues. Software today automatically creates a bill of materials (BoM) and tells you how many feet of wire would be required amongst other things. It needs to be three-dimensional and be able to compute distances that are not in a simple XY plane.

There might be companies today who do it all push-button, and then there are companies that do it semi-automatically, but no one today can still design a car manually. In fact, we even have motorcycle manufacturers who use this software.

What is one factor that could have the biggest impact on the EDA industry?

With most integrated circuits, the progressions of history have been designing them so that at most you could sell one chip per person or one chip per household and so on. With the Internet of Things (IoT), you go from selling millions or billions to selling tens of billions or hundreds of billions.

So when the number of designs that has to be done increases, the total revenue for the semiconductor industry increases. The information that has to be processed from the IoT requires massive resources for computing analysis.